

From: Sneller, Cindy (DEQ)
Sent: 11/15/2018 3:11:15 PM
To: "Doug Langworthy (dlangworthy@cadillac-MI.net)" <dlangworthy@cadillac-MI.net>
Subject: Land appl docs and DTs submitted for Biosolids AR - Questions
Attachments: should have picked this one.jpg

Hi Doug – I'm trying to figure out the land application documents you submitted for the Annual Report.

One is in an Excel spreadsheet and the others are pdfs of Ag rate calculations.

I cannot make heads or tails out of the Ag rate calculations. I'm trying to find on these sheets **how many Dry Tons (DT) of Biosolids were land applied on these fields** and I can't find that anywhere.

The one site you uploaded as an *Excel spreadsheet*, is nice because I can find the DT information.

Do you have sheets for each one of the other sites with the same info as the one (2018 As applied-29NS01)? Or, at least something for the other sites that tells me how many DTs of biosolids were applied to the site?

The other information is fine to submit, it's just that not all the information is on these sheets.

Also, the Applier from Iowa for this past Spring, on their certification form for Pathogen and Site Restrictions, they marked – 7 day fecal and Geometric mean. Cadillac doesn't do that one, you guys do Anaerobic Digestion. See my attachment, the process they should've marked.

To finish up this report review, I will need the following:

- A revised Pathogen & Site Restriction form from the Applier for the Spring haul.
- Documentation showing how many DTs per site were applied.

Then I can correctly figure out the DTs per your facility for the year. **Right now, I calculated 274 DTs and you reported 240.87 DTs.** That's a big difference.

Let me know if you have questions. I'm only in until 3:30 pm tomorrow (Fri. Nov. 16) and will be on vacation after that until Dec. 5, returning on that day at 7 am.

Cindy

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ATTACHMENT NAME:

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ATTACHMENT TYPE:

JPEG raster lossy image

Fecal Coliform geometric mean was <1 MPN / CFU per dry gram total solids for the reporting period.

Go to question 4

- 2) A Process to Significantly Reduce Pathogens (PSRP) was used to meet Class B pathogen reduction.

CHECK ALL PSRP USED TO MEET THE CLASS B PATHOGEN REDUCTION REQUIREMENT.

 Aerobic Digestion

Aerobic conditions were maintained for a mean cell residence time and temperature between 40 days at 20 C (68 F) or 60 days at 15 C (59 F).

 Air drying

Sewage sludge was dried on sand beds or on paved basins. The sewage sludge dried for a minimum of three (3) months. During two (2) of the three (3) months, the ambient average daily temperature was above 0 C (32 F).

 Anaerobic Digestion

Sewage sludge was treated in the absence of air for a specific mean cell residence time and temperature between 15 days at 35 to 55 C (95 to 131 F) or 60 days at 20 C (68 F).

 Composting

Using either the within-vessel, static aerated pile, or windrow composting methods, the temperature of the sewage sludge was raised to 40 C (104 F) or higher and remained at 40 C (104 F) or higher for five (5) days. For four (4) hours during the five (5) days, the temperature in the compost pile exceeded 55 C (131 F).

 Lime Stabilization